SapEM Systems
DASHBOARD GUIDE

SapEM Dashboard Guide

A complete, start-to-finish walkthrough of setting up your dashboard — from powering on the hub to placing every sensor, repeater, and tubing line on your map.

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1 Set Up the Receiving Hub

1. Install the antenna.
2. Plug in the HDMI cable to the board (either port) and desired display device.
3. Plug in powerblock and turn the switch on (if your version has one).
4. Plug in ethernet (recommended). If not using ethernet connect the hub to WiFi by clicking the WiFi symbol in the top right corner (mouse not included, can be installed in any USB port).

2 Create Your SapEM Account

1. On any device (besides the hub) go to www.concept-solid.com/sapem and click Login in the top right corner.
2. Click Create Account.
3. Follow directions on screen. Once completed, login to your dashboard.

3 Connect the Hub to Your Account

1. Make sure the hub is powered on and wait about 5 minutes to let it finish booting and connecting to the network.
2. On your dashboard, press the green "+ Add Hub" button in the top row.
3. In the "Add Receiving Hub" window, enter the **Hub Hardware ID** printed on the inside of the hub enclosure, and optionally give it a **Hub Name** (e.g. "North Sugarbush"). The name can be changed anytime.
4. Press "Add Hub." The dashboard will refresh and a new section for that hub will appear with its gauge, stats, and sensor table.

4 Add and Assign Sensors

1. Install the antenna on the sensor.
2. Power on the sensor by pressing the metal button on the side of the enclosure. Do the first binding close to the hub.

3 Wait about 30 seconds, then on the dashboard press the green "+ Add Sensor" button.

4 In the "Add Sensor Node" window, enter the 4-digit **Sensor Serial Number** printed inside the sensor enclosure, and optionally a **Sensor Name**. Press "Add Sensor." A brand-new sensor won't show readings immediately — data appears once it reports in, which can take a few minutes.

5 Press the "Assign Sensor" button at the top of the dashboard. Choose the sensor under **Select Sensor**, the hub under **Select Hub**, and pick an **Action**:

Assign to this hub — adds the sensor to the chosen hub without removing it from any other hub it's already on. Use this for a sensor that reports to more than one hub.

Move to this hub — makes the chosen hub the sensor's only hub, removing it from every other hub first.

Remove from this hub — unassigns the sensor from the chosen hub only.

6 Press "Apply." A sensor with no hub at all sits under the "Unassigned Sensors" table near the top of the dashboard until it's assigned.

It's recommended to charge all sensors fully before deploying for a season. The charging port is in the bottom right corner of the circuit board inside the box (USB-C cable included with every hub). While charging, the small indicator light on the board glows **red**; once fully charged it switches to **blue**. Wait about 30 seconds between powering on sensors so each one finishes binding before the next.

5 Reading Your Dashboard

1 Each hub gets its own card showing a live vacuum **gauge**, and **Hub Stats**: sensors on that hub, average temperature, average pressure, and when data was last received.

2 Under Hub Stats you can set an **alert threshold** — enter a percentage and any sensor reading that far below the hub's average pressure is flagged in a "Low pressure sensors" list and highlighted amber in the table.

3 Below that, each hub's sensor table lists every assigned sensor with its live P1/P2 pressure, temperature, signal strength, and last-seen time.

4 Tap any sensor's row (not the drag handle) to open a "spotlight" popup with just that sensor's gauge and readings. Tap the X, or tap outside the popup, to close it.

5 Press "↻ Refresh" at the top anytime to pull the latest readings and reconcile the map without reloading the page.

6 Renaming Things and Removing a Hub

1 Press the small pencil icon next to a hub's name to open "Edit Names."

2 Change the **Hub Name** at the top, and rename any of that hub's sensors in the list below it (each shows its serial number for reference). Press "Save" to apply all the changes at once.

3 To remove a hub entirely, press "Remove Hub" in that same window, then confirm. Any sensors assigned to it move to the "Unassigned Sensors" table rather than being deleted — reassign them to another hub whenever you like.

Removing a hub takes it off the dashboard completely. To bring it back later, use "+ Add Hub" again with the same Hub Hardware ID.

7 Reordering Sensors in a Hub

1 Each sensor row has a small grab handle (three pairs of dots) just to the left of its name.

2 Press and hold the handle, then drag the row up or down to the position you want — this works with a mouse or a finger on a phone.

3 Release to drop it in place. The new order is saved automatically for that hub and stays that way across refreshes.

8 Populating the Map

Scroll to the bottom of the dashboard to find the shared "Map" section — every hub and sensor you've added shows up here together on one satellite map of your sugarbush.

8a. Placing a Sensor

1. Any sensor without a map position yet shows up in a "Not yet placed on map" bar right above the map, with a "Place [sensor name]" button for each one.
2. Press that button (or, for a sensor already on the map, tap its dot and choose "Reposition marker" from the popup). A marker appears at the map's center and the map switches to placement mode: "Tap the map to reposition."
3. Tap the spot on the map where that sensor actually sits, or press "My Location" to jump the marker to your phone's current GPS position (then fine-tune with a tap if needed).
4. Press "Save" to lock in the position, or "Cancel" to back out. Saved sensors show as a colored dot on the map — the color reflects that sensor's current pressure reading.

8b. Adding Hubs, Repeaters, and Notes

Use freestanding markers for anything on the map that isn't a sensor — the hub itself, a signal repeater, or a plain note like "tree down."

1. In the top-left corner of the map, press the pink "Marker" button.
2. Tap the map to place it (or use "My Location"), then press "Next."
3. Choose what kind of marker it is — **Repeater**, **Hub**, or **Flag / Note** — each gets its own icon on the map.
4. Type a name (e.g. "Back Field Repeater") and press "Save."
5. Tap any freestanding marker later to Reposition, toggle its name label, or Delete it from its popup.

8c. Drawing Sap Tubing Lines

1. Press the "Lines" button (top-left of the map) to open the line panel.
2. If you have more than one hub, choose which hub the new line belongs to from the "Hub for new lines" dropdown.
3. Pick a line type: **Mainline**, **Lateral**, or **Dropline**.
4. Press "+ Draw Line," then tap the map once for each point along the tubing run. A running count shows how many points you've placed.
5. Press "Undo" to remove the last point, or "Finish" once you have at least two points to save the line (it's auto-named, e.g. "mainline 1"). "Cancel" discards the whole line.

8d. Editing, Labeling, and Deleting Lines

1. Tap anywhere on an existing line to select it. A small panel appears with Edit, Show/Hide Label, Delete, and Cancel.
2. **Edit** lets you drag any point to reshape the line, and drag or tap one of the small faded dots between points to add a new bend point there. You can also rename the line in the text field. Press "Save" or "Cancel."
3. **Show/Hide Label** keeps the line's name always visible on the map instead of only on tap.
4. **Delete** asks for confirmation, then removes the line permanently.

Sensor markers, freestanding markers, and lines each have their own "Show/Hide name (or label) on map" toggle — turn it on for anything you want visible at a glance without tapping it first.

9 Map Icon & Line Legend

- Sensor — good pressure
- Sensor — low pressure
- Sensor — critical / no reading
- Hub marker
- Repeater marker
- 📌 Flag / Note marker
- Mainline
- Lateral
- Dropline

Exact pressure-color thresholds may be tuned over time — the shapes (circle = sensor, square = hub, diamond = repeater, flag = note) are what stay consistent.